(iii) When SP = CP then shopkeeper is in no profit (or) no loss

$SP = CP \Rightarrow \frac{SP}{CP} = 1$

Ex:- Shopkeeper is selling @ C.P means $SP = CP \Rightarrow \frac{SP}{CP} = 1$ Means, No profit / No loss.

↳ In Profit & Loss

(iv) When two different S.P's are given by selling two identical articles @ same profit & loss, then

$S.P_1 = CP + P$
$S.P_2 = CP - L$ $\Rightarrow SP_1 + SP_2 = CP + CP$ [P=L]

$\Rightarrow 2CP = SP_1 + SP_2 \Rightarrow \boxed{CP = \frac{SP_1 + SP_2}{2}}$

(v) When two identicals articles of same cost price sold @ two different prices such a way that same profit on 1st, the same loss on 2nd. The net profit/loss on 2 articles is zero.

$SP_1 = CP + P$ P=L
$SP_2 = CP - L$

$SP_1 + SP_2 = CP + CP = 2CP$ [∵ No profit / No loss]

(vi) When 2 articles are sold @ same price [$SP_1 = SP_2$] such that Profit on 1st is same as loss on 2nd [P% = L% = P/L]. Then net gain / loss is always loss irrespective of what S.P's.

$\text{Net Loss \%} = \frac{[\text{Common Profit / Loss}]^2\%}{100} = \frac{[P^2 = L^2]\%}{100}$

(vii) When 2 (or) more successive discounts are given on a M.R.P one after the other [cumilative discounts how we have seen cumilative decreaments in %] x%, y%, z% ..... Then the discounted price Rs.

$$\text{discounted price} = \frac{(100-x)}{100}\frac{(100-y)}{100}\frac{(100-z)}{100}\ldots \times MRP$$

[viii] Formulae of Profit & C.P given SP = ?

$$\frac{SP}{CP} > 1 \Rightarrow \frac{SP}{CP} = 1 + P\% = 1 + \frac{P}{100}$$

$$\frac{SP}{CP} = \frac{100+P}{100} \Rightarrow \boxed{SP = CP \times \frac{(100+P)}{100}} \quad -(1)$$

[ix] Profit & SP given then CP = ?

from eqn (1) $\Rightarrow$

$$\boxed{CP = \frac{100 \times SP}{(100+P)}}$$

[x] When Loss $\Rightarrow$ Loss & CP given SP = ?

$$\frac{SP}{CP} < 1 \Rightarrow \frac{SP}{CP} = 1 - P\% = 1 - \frac{P}{100}$$

$$\Rightarrow \frac{SP}{CP} = \frac{(100-P)}{100} \Rightarrow \boxed{SP = CP \times \frac{(100-P)}{100}} \quad -(2)$$

[xi] When Loss $\Rightarrow$ Loss & SP given CP = ?

from eqn (2) $\rightarrow$ $SP = CP \times \frac{(100-P)}{100}$

$$\Rightarrow \boxed{CP = SP \times \frac{100}{(100-P)}}$$

(xii) When 'x' no. of articles S.P = 'y' no. of articles C.P

$\Rightarrow$ Profit / Loss % = $x \times SP = y \times CP$

$$\Rightarrow \frac{SP}{CP} = \left[\frac{y}{x} \times 100\right] \pm 100$$

when $[y > x] \Rightarrow$ it Profit % = $\left(\left[\frac{y}{x} \times 100\right] - 100\right)$ %

when $[y < x] \Rightarrow$ Loss % = $\left[100 - \left(\frac{y}{x} \times 100\right)\right]$ %

Ex:- 10 articles SP = 20 articles CP $\Rightarrow$ P/L % = ?

here $20 > 10 \Rightarrow y > x \Rightarrow$ profit.

$10 \times SP = 20 \times CP \Rightarrow \frac{SP}{CP} = \frac{20}{10} \Rightarrow [(2-1) \times 100 = P\%]$

$$P\% = \left[\left(\frac{20}{10} \times 100\right) - 100\right] = 100\%$$

Note:- Same with x % of SP = y % of CP given

(xiii) When shop keeper sells his goods @ cost price but he is giving x grams in place of 'y' grams then Profit / Loss % = ?

When $[y > x] \Rightarrow$ Profit & when $(y < x) \Rightarrow$ Loss

$$\text{Profit \%} = \frac{\text{Error } (y-x)}{x} \times 100 = \frac{\text{Error}}{\text{Original}} \times 100$$

$$\text{Loss \%} = \frac{\text{Error}(x-y)}{x} \times 100 = \frac{\text{Error}}{\text{Original}} \times 100$$

Ex:- Shopkeeper sells @ C.P but giving 750gms instead of 1000gms. Profit % = ?

Ans:- Profit is to be considered with cost price quantity.

here profit is 250g compared with 750 g.

$$P\% = \frac{250}{750} \times 100 = 33.33\%$$

(34)

Gvx) Shop keeper sells goods @ p% profit & giving x% instead of y%. The profit / Loss % = ?

↳ here Profit / Loss should be considered on 2 cases

1) According to money wise
2) According to weight wise } so,

$$\frac{SP}{CP} = \frac{SP_1}{CP_1} \times \frac{SP_2}{CP_2} \times \cdots\cdots$$

$\frac{SP_1}{CP_1}$ ↓ Money wise

$\frac{SP_2}{CP_2}$ ↓ weight wise

Ex:- Shop keeper sells his goods @ 20% profit & giving 800gms instead of 1000 gms. P/L % = ?

$$\frac{SP}{CP} = \left(\frac{SP_1}{CP_1} = 1.2 \text{ (20\% P)}\right) \times \left(\frac{SP_2}{CP_2} = \frac{\text{S.P Quantity}}{\text{C.P Quantity}} = \frac{1000}{800}\right)$$

$$= 1.2 \times \frac{1000}{800} = 1.2 \times 1.25 = 1.5$$

$$P\% = (1.5 - 1)100 = 50\% \uparrow$$

Discounts :- Of 3 types

(i) Flat discount

Ex:- M.R.P = 1000/-

Disc = 60%

SP ⇒ MPP = 1000

Dis = 60% of 1000 = -600

SP = 400

(ii) Successive disc.

Ex:- 1000/- = MRP

10% ↓, 20% ↓, 30% ↓

1000

−100 ⇒ 10% of 1000

900

−180 20% of 900

720

−216 30% of 720

SP = 504 (or)

SP = 0.9 × 0.8 × 0.7 × 1000

= 504

(iii) Equal Successive disc.

[20% + 20% + 20%]

Ex:- 1000 = MRP ⇒ 20% ↓, 20% ↓, 20% ↓

1000

−200 20% of 1000

800

−160 20% of 800

640

−128 20% of 640

512 = SP

SP = 0.8 × 0.8 × 0.8 × 1000

$= (0.8)^3 \times 1000$

$[SP = x^n \times MRP]$ → (1)